Modèles stochastiques en finance – Séance 2
Modèle d'un marché financier à un actif risqué.
Modèle de Black & Scholes.

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Table of contents

A single-asset financial market framework
- Model description
- No arbitrage and first fundamental theorem
- Completeness and second fundamental theorem

The Black & Scholes model
- Model description and properties
- The Black & Scholes formula
- Δ-hedging and the Black & Scholes PDE
- Greeks of the standard Black & Scholes formula
- Black & Scholes pros and cons

Table of contents

A single-asset financial market framework
- Model description
- No arbitrage and first fundamental theorem
- Completeness and second fundamental theorem

The Black & Scholes model
- Model description and properties
- The Black & Scholes formula
- Δ-hedging and the Black & Scholes PDE
- Greeks of the standard Black & Scholes formula
- Black & Scholes pros and cons

Table of contents

A single-asset financial market framework
- Model description
- No arbitrage and first fundamental theorem
- Completeness and second fundamental theorem

The Black & Scholes model
- Model description and properties
- The Black & Scholes formula
- Δ-hedging and the Black & Scholes PDE
- Greeks of the standard Black & Scholes formula
- Black & Scholes pros and cons

General single-asset financial market model

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space. Let $(B_t)_{t\in[0,T]}$ be a standard Brownian motion and let $(\mathcal{F}_t)_{t\in[0,T]}$ be its natural filtration. We consider a general financial market model in continuous-time with finite horizon $T > 0$. We assume that there exists a risk-free asset with price D satisfying the stochastic differential equation

$$dD_t = D_t r_t \, dt, \quad D_0 = 1,$$

where $(r_t)_{t\in[0,T]}$ is an adapted stochastic process. We assume that only one risky asset is traded and that its price S satisfies the stochastic differential equation

$$dS_t = \mu_t S_t \, dt + \sigma_t S_t \, dB_t, \quad S_0 = s_0 \in \mathbb{R},$$

where $(\mu_t)_{t\in[0,T]}$ is an adapted process and $(\sigma_t)_{t\in[0,T]}$ is an adapted process, $\mathbf{P}$-a.s. positive.

Asset prices and discounting

Proposition 1.1 (Asset prices)

(a) *The price of the risk-free asset at time* $t \in [0, T]$ *is* $D_t = e^{\int_0^t r_s \, ds}$.

(b) *The price of the risky asset at time* $t \in [0, T]$ *is*

$$S_t = s_0 \exp\left[\int_0^t (\mu_s - \frac{\sigma_s^2}{2}) \, ds + \int_0^t \sigma_s \, dB_s\right].$$

- This section defines a general formal framework. Conditions apply on r_t, μ_t and σ_t to apply stochastic calculus results.

Proposition 1.2 (Discounting)

If $(X_t)_{t \in [0,T]}$ *is a price/value process in the model,* $\tilde{X}_t = D_t^{-1} X_t$ *is the discounted price/value process.*

Self-financing investment strategies

- We assume that *at any time* t an investor holds $\Delta_t \in \mathbb{R}$ shares of the risky asset and that the rest of his wealth is invested in the risk-free asset.
- Let V_t be the value of the portfolio at time t.

Définition 1.3 (Self-financing portfolios/strategies)

A portfolio is self-financing if the variations of its value are exclusively due to the variations of asset prices, i.e. if the process $(V_t)_{t\geq 0}$ *satisfies*

$$dV_t = \Delta_t \, dS_t + (V_t - \Delta_t S_t) r_t \, dt.$$

- Self-financing : ni cash deposit or withdrawal.
- Strategy $(\Delta_t)_{t\in[0,T]}$ - portfolio $(V_t)_{t\in[0,T]}$.
- Rewriting the self-financing condition :

$$dV_t = r_t V_t \, dt + \Delta_t S_t (\mu_t - r_t) \, dt + \Delta_t S_t \sigma_t \, dB_t.$$

Equivalent martingale measure

Proposition 1.4 (Equivalent martingale measure)

(a) *With some conditions, there exists a measure* $\mathbf{Q}$ *equivalent to* $\mathbf{P}$ *and such that*

$$d\tilde{S}_t = \sigma_t \tilde{S}_t \, dB_t^{\mathbf{Q}},$$

where $B^{\mathbf{Q}}$ *is a* $\mathbf{Q}$*-Brownian motion.*

(b) *If* $\tilde{V}$ *is the discounted value of a self-financing portfolio holding at time* t *the quantity* Δ_t *of risky assets, then*

$$d\tilde{V}_t = \sigma_t \Delta_t \tilde{S}_t \, dB_t^{\mathbf{Q}}.$$

- $\mathbf{Q}$ is called an equivalent martingale measure.
- Suitable integrability conditions need to be satisfied.

Pricing of a replicable derivative product

Définition 1.5 (Replication of derivatives products)

Let us consider a derivative product with an $\mathcal{F}_T$-measurable payoff H. We say that this product can be replicated if there exists a self-financing portfolio with value $(V_t)_{t\in[0,T]}$ such that $V_T = H$ $\mathbf{P}$-a.s.

Proposition 1.6 (Pricing of a replicable derivative product)

If V_t is the value at time t of a self-financing portfolio replicating some $\mathcal{F}_T$-measurable payoff H, then

$$V_t = \mathbf{E}^{\mathbf{Q}}\left[e^{-\int_t^T r_s\,ds} H \mid \mathcal{F}_t\right], \ t \in [0, T].$$

- This is the martingale approach to pricing in finance.
- Suitable integrability conditions need to be satisfied.

Table of contents

A single-asset financial market framework
- Model description
- No arbitrage and first fundamental theorem
- Completeness and second fundamental theorem

The Black & Scholes model
- Model description and properties
- The Black & Scholes formula
- Δ-hedging and the Black & Scholes PDE
- Greeks of the standard Black & Scholes formula
- Black & Scholes pros and cons

No arbitrage and the law of one price

Définition 1.7 (Arbitrage)

An investment strategy is called an arbitrage opportunity if it is a self-financing strategy such that the value $(V_t)_{t\in[0,T]}$ of the associated portfolio value satisfies :

$$V_0 = 0, \quad \mathbf{P}(V_T \geq 0) = 1, \text{ and } \mathbf{P}(V_T > 0) > 0.$$

- If there exists an arbitrage on $[0, t^*]$, $t^* \in]0, T]$, then there exists an arbitrage on $[0, T]$.
- Assumption of no arbitrage : fundamental assumption in classic economics/finance ; linked to the equilibrium of markets.

Proposition 1.8 (Law of one price)

If there is no arbitrage, and if two self-financing portfolios with values V^1 and V^2 are such that $V_T^1 = V_T^2$ $\mathbf{P}$-a.s., then for all $t \in [0, T]$, $V_t^1 = V_t^2$ $\mathbf{P}$-a.s.

First fundamental theorem of arbitrage pricing

Theorem 1.9 (First fundamental theorem of arbitrage pricing)

If a financial market model admits an equivalent martingale measure, then there is no arbitrage opportunity in this model.

- The above statement is an implication, not an equivalence. Equivalence is obtained in discrete time with finite state space models (see FMD). But things are more involved in continuous time. Modified definitions of arbitrage (free lunch with vanishing risk) and more complex technical results are needed in continuous time to get an equivalence.

Put-call parity

Proposition 1.10 (Put-call parity)

Let us consider a financial market model with an equivalent martingale measure $\mathbf{Q}$*. Let* C_t *(resp.* P_t*) be the price of the standard European call (resp. put) option with strike* K *and maturity* T*. Then for any* $t \in [0, T]$*,*

$$C_t - P_t = S_t - K\, Z(t, T),$$

where $Z(t, T)$ *is the price at time* t *of the zero-coupon bond with maturity* T *(see the Fixed Income course).*

- In the case of a constant rate $r > 0$ (see Black & Scholes next section), then $Z(t, T) = e^{-r(T-t)}$.
- In the general case, $Z(t, T) = \mathbf{E}^{\mathbf{Q}}\left[e^{-\int_t^T r_s\, ds} \mid \mathcal{F}_t\right]$ (see the Fixed Income course).

Table of contents

A single-asset financial market framework
- Model description
- No arbitrage and first fundamental theorem
- Completeness and second fundamental theorem

The Black & Scholes model
- Model description and properties
- The Black & Scholes formula
- Δ-hedging and the Black & Scholes PDE
- Greeks of the standard Black & Scholes formula
- Black & Scholes pros and cons

Martingale representation theorem

Theorem 1.11 (Martingale representation theorem)

Let $T > 0$ be a finite horizon. Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space. Let B be a Brownian motion and $(\mathcal{F}_t)_{t\in[0,T]}$ be its natural filtration. If $(M_t)_{t\in[0,T]}$ is square-integrable $\mathcal{F}_t$-martingale, then there exists a $\mathcal{F}_t$-adapted process $(\alpha_t)_{t\in[0,T]}$ satisfying $\mathbf{E}\left[\int_0^T \alpha_s^2\, ds\right] < \infty$, and such that

$$\forall t \in [0, T], M_t = M_0 + \int_0^t \alpha_s\, dB_s \quad \mathbf{P} - a.s.$$

- Converse of the martingale property of the stochastic integral with respect to the Brownian motion.

Second fundamental theorem of arbitrage pricing

Définition 1.12 (Complete markets)

A financial market model is said to be complete if any derivative product with an $\mathcal{F}_T$-measurable payoff can be replicated.

Theorem 1.13 (Second fundamental theorem of arbitrage pricing)

Let us consider a single-asset financial market model driven by a single Brownian motion. If the model admits an equivalent martingale measure, then it is complete and the equivalent martingale measure is unique.

Table of contents

A single-asset financial market framework
- Model description
- No arbitrage and first fundamental theorem
- Completeness and second fundamental theorem

The Black & Scholes model
- Model description and properties
- The Black & Scholes formula
- Δ-hedging and the Black & Scholes PDE
- Greeks of the standard Black & Scholes formula
- Black & Scholes pros and cons

Table of contents

A single-asset financial market framework
- Model description
- No arbitrage and first fundamental theorem
- Completeness and second fundamental theorem

The Black & Scholes model
- Model description and properties
- The Black & Scholes formula
- Δ-hedging and the Black & Scholes PDE
- Greeks of the standard Black & Scholes formula
- Black & Scholes pros and cons

Model assumptions

- Time-continuous financial model with fixed finite horizon $T > 0$.
- $(\Omega, \mathcal{F}, \mathbf{P})$ probability space with filtration $(\mathcal{F}_t)_{t\in[0,T]}$.
- B $(\mathcal{F}_t)$-adapted standard one-dimensional Brownian motion
- A risk-free asset with deterministic price D, continuously accruing with continuous risk-free rate $r > 0$:
$$dD_t = rD_t\,dt, D_0 = 1.$$
- A single risky asset with stochastic price S solution of the stochastic differential equation
$$dS_t = \mu S_t\,dt + \sigma S_t\,dB_t,$$
with $\mu \in \mathbb{R}$ constant and $\sigma \in \mathbb{R}_+^*$ positive constant.

Asset prices in the Black & Scholes model

Proposition 1.14 (Explicit forms)

(a) $\forall t \in [0, T], D_t = e^{rt}$;

(b) $\forall t \in [0, T], S_t = S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma B_t}$.

Proposition 1.15 (Properties of the price of the risky asset)

(a) $(S_t)_{t \in [0,T]}$ *is a continuous process.*

(b) *Far all* $t \in [0, T]$, S_t *is a log-Gaussian random variable.*

(c) *Asset returns are stationary and independent.*

No arbitrage and martingale pricing

- The Black & Scholes model is a specific case of the general formal framework of the previous section with constant risk-free rate, constant drift and constant volatility of the risky asset price.
- The Black & Scholes model admits an equivalent martingale measure $\mathbf{Q}$ defined with the Girsanov theorem by $\frac{d\mathbf{Q}}{d\mathbf{P}} = \exp\left(-\frac{\mu - r}{\sigma} B_T - \left(\frac{\mu - r}{\sigma}\right)^2 T\right)$.
- $S_t = S_0 e^{\left(r - \frac{\sigma^2}{2}\right)t + \sigma B_t^{\mathbf{Q}}}$ with $B^{\mathbf{Q}}$ $\mathbf{Q}$-Brownian motion.
- The Black & Scholes model has no arbitrage opportunities.
- The Black & Scholes model is complete.
- In the Black & Scholes model, the price of a derivative product with $\mathcal{F}_T$-measurable payoff H is at time $t \in [0, T]$

$$V_t = \mathbf{E}^{\mathbf{Q}}\left[e^{-r(T-t)} H \mid \mathcal{F}_t\right].$$

Table of contents

A single-asset financial market framework
- Model description
- No arbitrage and first fundamental theorem
- Completeness and second fundamental theorem

The Black & Scholes model
- Model description and properties
- The Black & Scholes formula
- Δ-hedging and the Black & Scholes PDE
- Greeks of the standard Black & Scholes formula
- Black & Scholes pros and cons

The Black & Scholes formula for vanilla call options

Proposition 1.16 (Black & Scholes formula for vanilla call options)

In the Black & Scholes model, the price C_t at time $t \in [0, T]$ of a European call option on the underlying S with maturity T and exercise price (strike) K is :

$$C_t = S_t \Phi(d_1) - K e^{-r(T-t)} \Phi(d_2),$$

where Φ is the standard Gaussian cdf and :

$$d_1 = \frac{\ln\left(\frac{S_t}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}},$$

$$d_2 = \frac{\ln\left(\frac{S_t}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}} = d_1 - \sigma\sqrt{T-t}.$$

Other forms of Black & Scholes-type formulas I

- Black & Scholes formula for vanilla put options
  In the Black & Scholes model, the price P_t at time $t \in [0, T]$ of a European put option on the underlying S with maturity T and exercise price (strike) K is :

$$P_t = Ke^{-r(T-t)}\Phi(-d_2) - S_t\Phi(-d_1),$$

  using d_1 and d_2 defined above. This formula can be obtained by direct computation, or via the call-put parity and the Black & Scholes call formula.

Other forms of Black & Scholes-type formulas II

- Black & Scholes formula with a continous dividend rate
  Assume that the stock pays the owner of a share a dividend, continuously, with known rate $a > 0$ per time unit. In this modified Black & Scholes model, the price C_t a European call option becomes :

$$C_t = S_t e^{-a(T-t)}\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2),$$

with $d_1 = \dfrac{\ln\left(\frac{S_t}{K}\right) + \left(r - a + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}$ and $d_2 = d_1 - \sigma\sqrt{T-t}$.

Other forms of Black & Scholes-type formulas III

- Black & Scholes formula with discrete dividends
  Assume that the stock pays the owner of a share a dividend with known rate $a_i > 0$ at time deterministic t_i, $0 < t_1 < \ldots < t_n < T$. In this modified Black & Scholes model, the price C_0 a European call option at time 0 becomes :

$$C_0 = S_0 \prod_{i=1}^{n} (1 - a_i)\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2),$$

with $d_1 = \dfrac{\ln\left(\frac{S_t}{K}\right) + \sum_{i=1}^{n} \log(1 - a_i) + \left(r + \frac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}}$ and $d_2 = d_1 - \sigma\sqrt{T}$.

Other forms of Black & Scholes-type formulas IV

- Black & Scholes formula with deterministic volatility function
  Assume that the volatility of the risky asset is a deterministc function $\sigma(t)$. In this modified Black & Scholes model, the price C_t of a call option is :

$$C_t = S_t \Phi(d_1) - K e^{-r(T-t)} \Phi(d_2),$$

with $d_1 = \dfrac{\ln\left(\frac{S_t}{K}\right) + \left(r + \frac{\bar{\sigma}_{t,T}^2}{2}\right)(T-t)}{\bar{\sigma}_{t,T}\sqrt{T-t}}$, $d_2 = d_1 - \bar{\sigma}_{t,T}\sqrt{T-t}$ and

$$\bar{\sigma}_{t,T} = \sqrt{\frac{1}{T-t}\int_t^T \sigma^2(s)\, ds}.$$

- ... many Black & Scholes formulas with practitioners' favorite tweaks (repo rates, etc.)

Table of contents

A single-asset financial market framework
- Model description
- No arbitrage and first fundamental theorem
- Completeness and second fundamental theorem

The Black & Scholes model
- Model description and properties
- The Black & Scholes formula
- Δ-hedging and the Black & Scholes PDE
- Greeks of the standard Black & Scholes formula
- Black & Scholes pros and cons

The Black & Scholes PDE

Proposition 1.17 (The Black & Scholes PDE)

Let us consider a derivative product on the underlying asset S, with maturity T and payoff $H = h(S_T)$, with $h : \mathbb{R}_+ \to \mathbb{R}_+$. In the Black & Scholes model, the price $v(t, x)$ of this derivative product at time $t \in [0, T]$ given $S_t = x$ satisfies the following partial differential equation :

$$\frac{\partial v}{\partial t}(t,x) + rx\frac{\partial v}{\partial x}(t,x) + \frac{1}{2}\sigma^2 x^2 \frac{\partial^2 v}{\partial x^2}(t,x) = rv(t,x), \quad t \in [0,T], x \in \mathbb{R}_+,$$

with terminal condition : $v(T, x) = h(x)$, $x \in \mathbb{R}_+$.

- The Black & Scholes PDE does not depend on the (European) derivative product. The payoff impacts only the terminal condition.
- This equation can be transformed into an homogenous heat equation. By solving it one retrieves the Black & Scholes formula.

Δ-hedging in the Black & Scholes model

- Within the Black & Scholes model,

$$\Delta_t = \frac{\partial v}{\partial x}(t, S_t)$$

  defines a strategy that replicates exactly the value of the derivative product : this is the Δ-hedging strategy.
- In more complex models (later in the course), this is not true. Δ-hedge still refers to $\Delta_t = \frac{\partial v}{\partial x}(t, S_t)$, but this is not an exact replication anymore.

Table of contents

A single-asset financial market framework
- Model description
- No arbitrage and first fundamental theorem
- Completeness and second fundamental theorem

The Black & Scholes model
- Model description and properties
- The Black & Scholes formula
- Δ-hedging and the Black & Scholes PDE
- Greeks of the standard Black & Scholes formula
- Black & Scholes pros and cons

Greeks of the standard Black & Scholes formula I

We outline the call price dependency to various parameters by writing $C_t = c(t, S, r, \sigma)$. Sensitivities of the option price to these parameters are named "Greeks".

Proposition 1.18

(a) *("delta")* $\Delta = \frac{\partial c}{\partial S} = \Phi(d_1) > 0$;

(b) *("gamma")* $\Gamma = \frac{\partial^2 c}{\partial S^2} = \frac{\Phi'(d_1)}{S\sigma\sqrt{T-t}} > 0$;

(c) *("theta")* $\Theta = \frac{\partial c}{\partial t} = \frac{-\sigma S}{2\sqrt{T-t}}\Phi'(d_1) - rKe^{-r(T-t)}\Phi(d_2) < 0$;

(d) *("vega")* $\nu = \frac{\partial c}{\partial \sigma} = S\sqrt{T-t}\Phi'(d_1) > 0$;

(e) *("rho")* $\rho = \frac{\partial c}{\partial r} = K(T-t)e^{-r(T-t)}\Phi(d_2) > 0$.

- Proof : Start by verifying that $S_t\Phi'(d_1) = Ke^{-r(T-t)}\Phi'(d_2)$.

Greeks of the standard Black & Scholes formula II

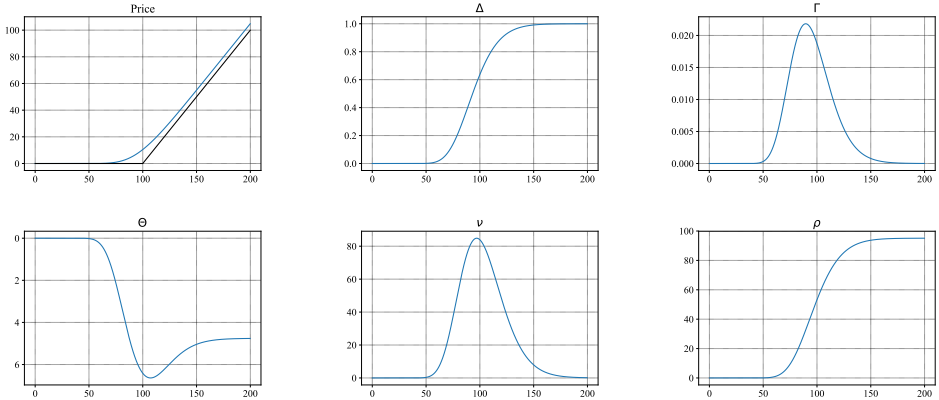

Figure – Black & Scholes price, $\Delta, \Gamma, \Theta, \nu$ and ρ of the standard call option, as functions of the spot price S_0 for the following parameters values : $r = 0.05$, $\sigma = 0.20$, $K = 100$ and $T = 1$.

Quick questions

(a) What is the probability (using the martingale measure) of exercising a call ?

(b) In the case of small interest rates / high volatility, what are zero-th and first order approximations of the ATM call delta w.r.t. the time to maturity $\tau = T - t$?

(c) Can an investor with zero initial capital sell a call option and Δ-hedge this short position without borrowing cash ?

(d) Why is a long call Δ-hedged position called long-Γ ?

Black & Scholes delta-hedging in "a" real world

- Assume that the *true* dynamics of the underlying asset under the martingale measure is given by the stochastic differential equation

$$dS_t = rS_t\,dt + \beta_t S_t\,dB_t.$$

- Assume that a bank sells to one of its client a standard call option on the underlying stock S, with strike K and maturity T. Assume also that the bank trader prices the option using the Black & Scholes formula and then hedges his portfolio with the replicating strategy $\Delta_t = \frac{\partial C}{\partial S}(t, S_t)$ with initial capital $V_0 = C(0, S_0)$. Let V_t be the value of the hedging portfolio.
- Then the hedging error $\xi_t = V_t - C(t, S_t)$ is at maturity :

$$\xi_T = \int_0^T \frac{1}{2} e^{r(T-t)} \left(\sigma^2 - \beta_t^2\right) \Gamma(t, S_t) S_t^2\,dt,$$

where $\Gamma(t, S_t) = \frac{\partial^2 C}{\partial S^2}(t, S_t)$. *(Proof later in the course).*

Table of contents

A single-asset financial market framework
- Model description
- No arbitrage and first fundamental theorem
- Completeness and second fundamental theorem

The Black & Scholes model
- Model description and properties
- The Black & Scholes formula
- Δ-hedging and the Black & Scholes PDE
- Greeks of the standard Black & Scholes formula
- Black & Scholes pros and cons

Black & Scholes pros and cons

- Explicit formula for the vanilla option prices.
- Explicit hedging strategy, analytical tools for hedging and risk management.
- Fundamental building block for a trading industry on derivatives products.
- Yet obtained with strong/restrictive assumptions :
  - log-normal asset dynamics ;
  - constant rate r, constant volatility σ ;
  - unique rate for lending and borrowing ;
  - continuous trading (no open/close) ;
  - perfect liquidity ;
  - infinitely divisible assets ;
  - short selling authorized ;
  - no transaction cost ;
  - no taxation.